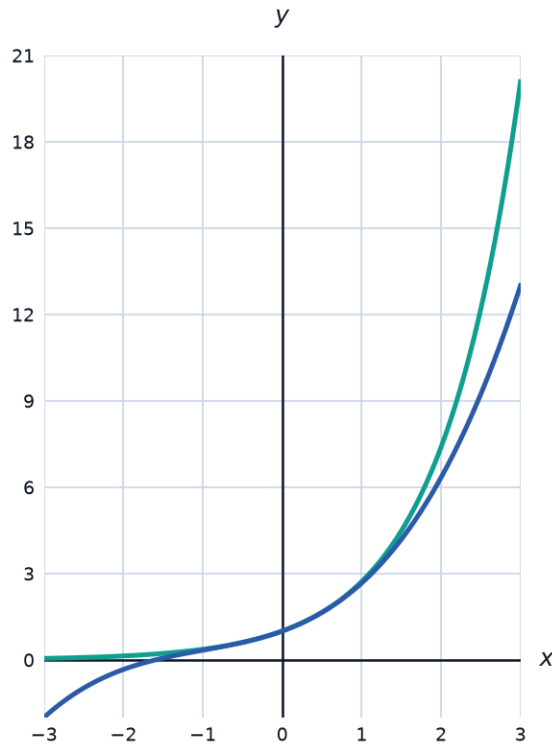


# Maclaurin Series & Convergence of Series

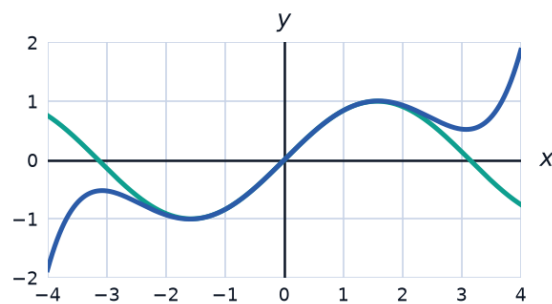


## Constructing a Maclaurin series

- 1 Find the first four non-zero terms of the Maclaurin series for  $f(x) = e^x$ , using  $f(x) = f(0) + f'(0)x + (f''(0)/2!)x^2 + (f'''(0)/3!)x^3 + \dots$ .



- 2 Find the Maclaurin series for  $f(x) = \sin x$  up to the term in  $x^5$ .



## Using known series to build new ones

- 3 Using the Maclaurin series for  $e^x$ , find the Maclaurin series for  $e^{-2x}$  up to the term in  $x^3$ .
- 4 Given the Maclaurin series  $\sin x \approx x - x^3/6$  and  $\cos x \approx 1 - x^2/2$ , find the Maclaurin series for  $\tan x$  up to the term in  $x^3$ , using  $\tan x = \sin x / (\cos x)$  and polynomial division.

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## Using a Maclaurin series to approximate a value

- 5 Use the Maclaurin series for  $e^x$  up to the  $x^3$  term to approximate  $e^{0.1}$ , to four decimal places, and compare with the true value  $e^{0.1} \approx 1.10517$ .
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## Tests for convergence of a series

- 6 Use the ratio test to determine whether the series  $1/2 + 2/4 + 3/8 + 4/16 + \dots + n/2^n + \dots$  converges.
- 7 Use the ratio test to determine whether the Maclaurin series for  $e^x$ ,  $1 + x + x^2/2! + x^3/3! + \dots + x^n/n! + \dots$ , converges for all real  $x$ .
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## Determining the radius of convergence

- 8 Use the ratio test to find the radius of convergence of the power series  $x + x^2/2 + x^3/3 + \dots + x^n/n + \dots$ .
- 9 Find the radius of convergence of the series  $1 + x + 2x^2 + 6x^3 + \dots + n! x^n + \dots$ , and explain what this tells us about where the series is useful.
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## Comparing a Maclaurin approximation to the exact function

- 10 Using the Maclaurin series  $\cos x \approx 1 - x^2/2 + x^4/24$ , approximate  $\cos(0.3)$  to four decimal places, and compare with the true value  $\cos(0.3) \approx 0.9553$ .
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## Maclaurin series for $\ln(1+x)$

- 11 Find the Maclaurin series for  $f(x) = \ln(1+x)$  up to the term in  $x^4$ , using  $f(x) = f(0) + f'(0)x + (f''(0)/2!)x^2 + (f'''(0)/3!)x^3 + (f^{(4)}(0)/4!)x^4$ .
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## Determining the interval of convergence using the ratio test

- 12 Find the radius of convergence of the Maclaurin series for  $\ln(1+x)$ ,  $x - x^2/2 + x^3/3 - \dots + (-1)^{n+1}x^n/n + \dots$ , using the ratio test.
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## Combining Maclaurin series through multiplication

- 13 Using the Maclaurin series  $e^x \approx 1 + x + x^2/2$  and  $1/(1-x) \approx 1 + x + x^2$  (valid for small  $x$ ), find the Maclaurin series for  $e^x/(1-x)$  up to the term in  $x^2$ , by multiplying the two series together.
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## Using differentiation to find a Maclaurin series

- 14 Given the Maclaurin series  $1/(1-x) \approx 1 + x + x^2 + x^3 + \dots$  for  $|x| < 1$ , differentiate both sides term by term to find the Maclaurin series for  $1/(1-x)^2$  up to the term in  $x^2$ .
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## Estimating an error bound in a Maclaurin approximation

- 15 The Maclaurin series for  $\sin x$  truncated after the  $x^3$  term is  $\sin x \approx x - x^3/6$ . Use this to approximate  $\sin(0.2)$ , and find the absolute error compared to the true value  $\sin(0.2) = 0.1986693\dots$
- 16 Find the first three non-zero terms of the Maclaurin series for  $f(x) = (1+x)^{1/2}$ , using the binomial series  $(1+x)^n \approx 1 + nx + n(n-1)/2x^2$  with  $n = 1/2$ .
- 17 Use the Maclaurin series for  $e^x$  up to the  $x^4$  term to approximate  $\lim_{x \rightarrow 0} (e^x - 1 - x)/x^2$ , and check that your answer matches the value obtained using L'Hopital's rule.
- 18 This question has three parts. (a) Find the Maclaurin series for  $f(x) = \cos(x^2)$  up to the term in  $x^8$ , by substituting  $x^2$  into the known series  $\cos u \approx 1 - u^2/2 + u^4/24$ . (b) Use your result to write down an approximation for  $\int_0^1 \cos(x^2) dx$  as a sum of simple fractions (this integral has no elementary closed form, so series approximation is genuinely useful here). (c) Evaluate this approximation as a single decimal, to three decimal places.
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## Checking endpoints of an interval of convergence

- 19 The Maclaurin series for  $\ln(1+x)$  has radius of convergence  $R = 1$  (found earlier in this worksheet). Explain why checking the endpoints  $x = 1$  and  $x = -1$  separately is necessary to determine the full interval of convergence, given that at  $x = 1$  the series becomes  $1 - 1/2 + 1/3 - 1/4 + \dots$  (the alternating harmonic series, which is known to converge).
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## Using a Maclaurin series to evaluate a standard trigonometric limit

- 20 Use the Maclaurin series for  $\sin x$  up to the  $x^3$  term to evaluate  $\lim_{x \rightarrow 0} (\sin x)/x$ .
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## Identifying the general term of a Maclaurin series

21 Which of the following is the general term of the Maclaurin series for  $\cos x$ ?

- a.  $(-1)^n x^{2n}/(2n)!$
- b.  $(-1)^n x^n/n!$
- c.  $x^{2n}/(2n)!$
- d.  $(-1)^{n+1} x^{2n-1}/(2n-1)!$

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## Comparing the accuracy of two Maclaurin approximations

22 Both  $e^x \approx 1 + x + x^2/2$  and  $\cos x \approx 1 - x^2/2$  are second-order Maclaurin approximations. Without calculating exact values, explain which approximation is likely more accurate at  $x = 1$ , referencing the size of the next omitted term in each series.